

4W'S INUA JAMII FOUNDATION — WEBSITE & SYSTEM INFRASTRUCTURE SUMMARY

The system is designed in a way that allows the foundation to start professionally while keeping operational costs manageable and scalable as the organization grows.

DOMAIN & BUSINESS EMAIL SYSTEM

We will use **Hostinger** for the domain and business email setup.

This package is favorable because it provides:

- Professional organizational emails
- Domain management
- Email infrastructure
- Affordable pricing

Estimated Cost:

- Approximately **KSh 3,250 – 3,500** (*This package holds us for 1yr*)

The package includes:

- 2 real email mailboxes
- 10 email aliases
- Around 1000 outgoing emails/day

Important distinction:

- A mailbox is a real email account with its own inbox and password.
- An alias is simply another email name that forwards messages to an existing mailbox.

Example:

membership@4wsfoundation.org can forward to support@4wsfoundation.org without needing another paid inbox.

This helps us appear organized and professional while reducing unnecessary costs.

EMAIL AUTOMATION SYSTEM

We will use SendGrid Free Tier for automated emails.

This system will handle:

- Welcome emails
- Password resets
- Membership confirmations
- Event confirmations
- Verification emails
- Automated notifications

The free plan supports:

- Around ***100 emails/day***

This is sufficient for the beginning phase considering we expect around 500 users initially.

Email will be our PRIMARY communication channel because it is:

- More affordable
- Easier to scale

- Better for bulk communication

As the organization grows and communication increases, we can scale to SendGrid Pro.

Estimated future upgrade:

- Approximately ***KSh 2,000 – 3,000/month***

BACKEND SYSTEM (DATABASE & SYSTEM LOGIC)

We will use ***Supabase Free Tier*** for the backend infrastructure.

The backend acts as the “brain” of the platform and handles:

- Member information
- Authentication/login
- CMS management
- Events
- QR membership verification
- Notifications
- Media storage
- System automation

The free tier can comfortably support:

- Hundreds to thousands of registered users
- Around ***50 – 200 concurrent active users***

(Concurrent users means people actively using the system at the same exact time.)

We will only scale when:

- Storage begins filling up
- Media files increase significantly
- Traffic grows heavily
- System operations become more demanding

Estimated future upgrade:

- Approximately ***KSh 3,200 – 3,500/month***

FRONTEND WEBSITE HOSTING

We will use ***Vercel Free Tier*** for hosting the website frontend.

This platform is excellent for:

- Fast performance
- Modern websites
- Reliability
- Smooth user experience

The free tier can comfortably support:

- Around ***50 – 300 concurrent users***

This is sufficient for the beginning phases because the platform is mainly:

- Membership-based
- Content-based

- Event/community focused

We will only scale if traffic and system usage increase significantly.

M-PESA / DARAJA API

We plan to implement Safaricom Daraja API for:

- Membership payments
- Donations
- Event payments

The API integration itself is **FREE**.

Only normal M-PESA transaction charges apply during payments.

SMS SYSTEM (LATER IMPLEMENTATION)

We plan to use Africa's Talking for SMS communication.

SMS will mainly be used for:

- Urgent alerts
- Important reminders
- Critical notifications

This is because SMS comes with recurring operational costs.

Estimated SMS cost:

- Around ***KSh 1 per SMS***

Example:

Sending one SMS to 500 users would cost approximately:

- Around ***KSh 500+***

To help reduce operational costs:

- Email will remain the main communication channel
- SMS will only be used where necessary

OVERALL STRATEGY

This infrastructure approach allows the foundation to:

- Start professionally
- Keep costs manageable
- Build a scalable system
- Grow gradually with the community
- Avoid unnecessary early expenses

The goal is to have a strong digital foundation that can grow together with the organization over time.